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- 5 Meselson and Stahl provided data that supported the theory that DNA replicates by semi-conservative replication.

(a) State the meaning of the term semi-conservative replication.

(2)

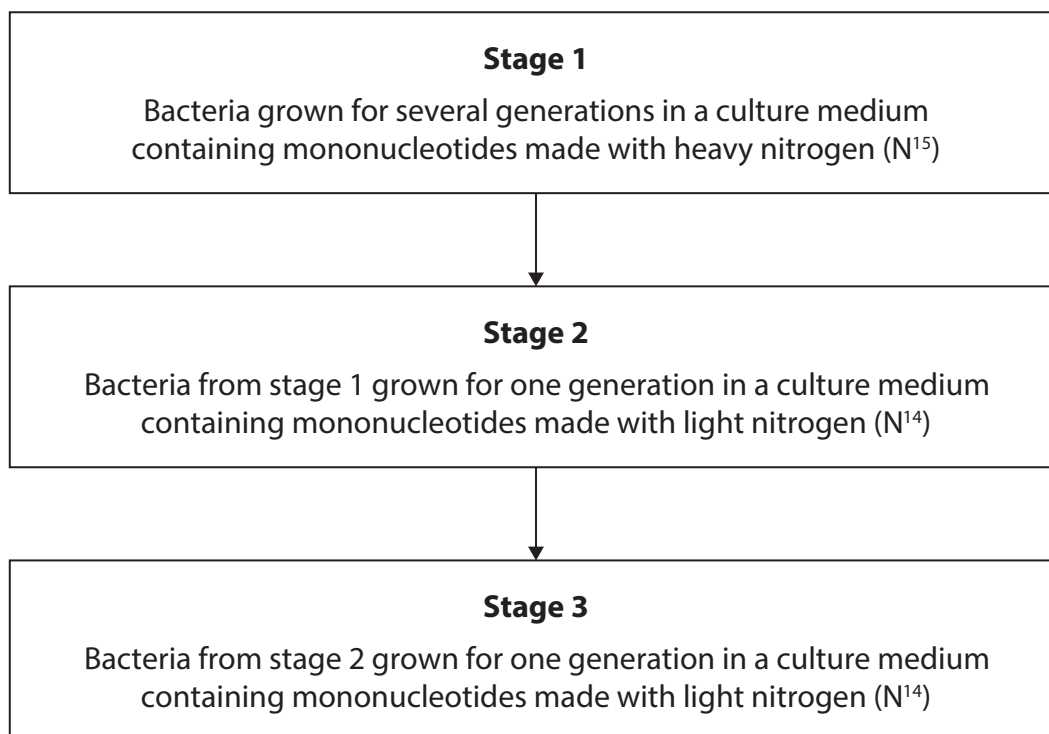
(b) Describe the role of DNA polymerase in the process of DNA replication.

(2)



- (c) Meselson and Stahl used mononucleotides made with either heavy nitrogen (N^{15}) or light nitrogen (N^{14}) in their experiments.

The flowchart summarises part of one experiment performed by Meselson and Stahl.

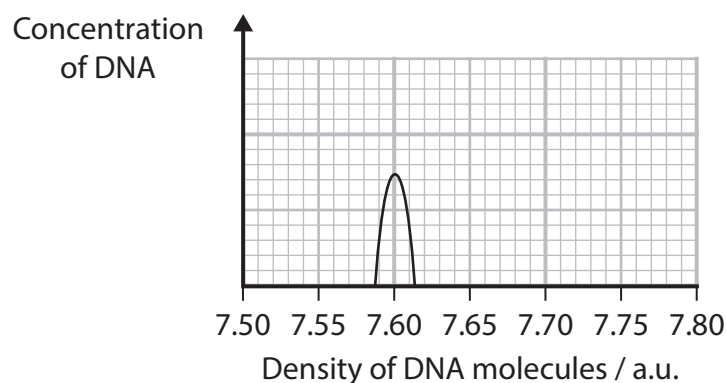


After each stage, a sample of DNA was taken from the bacteria and the DNA molecules separated on a density gradient in a tube.

- DNA with only heavy nitrogen has a mean density of 7.74 a.u.
- DNA with only light nitrogen has a mean density of 7.60 a.u.

Meselson and Stahl measured the concentration of DNA with different densities.

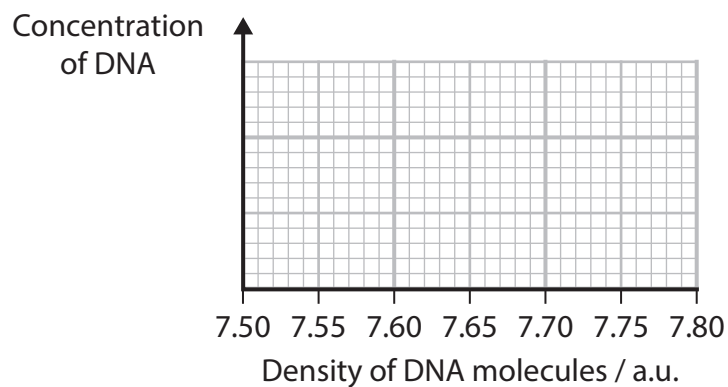
The graph shows the results of these measurements for DNA molecules with only light nitrogen.



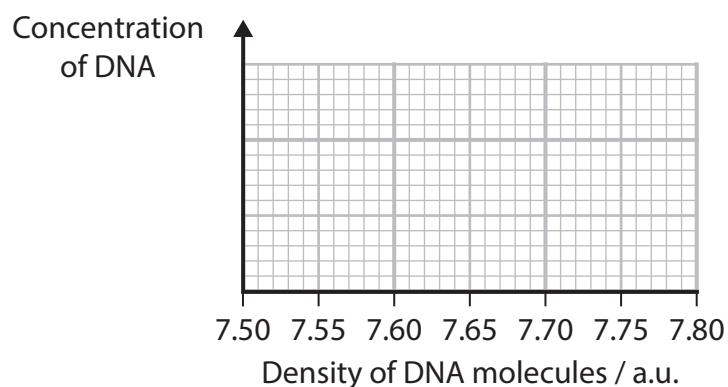
Complete the graphs to show the concentrations and densities of the DNA molecules produced at the end of stages 1, 2 and 3 shown in the flowchart.

(6)

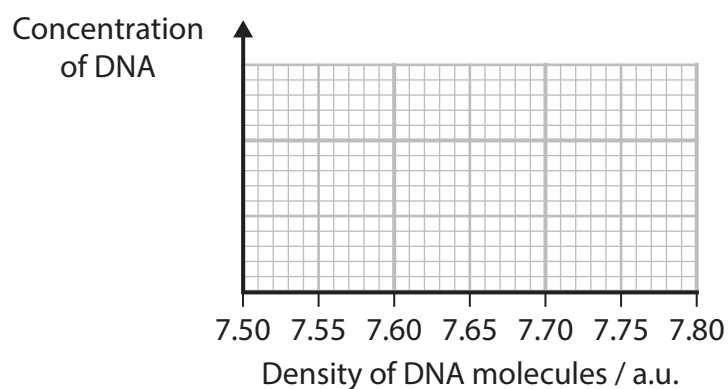
End of stage 1:



End of stage 2:



End of stage 3:



(Total for Question 5 = 10 marks)